

Chapter 24 - Executable Knowledge: Stage 7 of Semantic Knowledge Development Lifecycle

From Intelligent Knowledge to Actionable Systems

The first six stages of the Semantic Knowledge Development Lifecycle (SKDL) transformed an abstract conceptual vocabulary into a **validated, intelligent, machine-reasonable** knowledge asset.

- Conceptual Modeling gave us structure.
- Semantic Description gave us meaning.
- Knowledge Reuse gave us composability.
- Semantic Governance gave us discipline.
- Semantic Validation gave us trust.
- Semantic Reasoning gave us intelligence.

Now, as knowledge begins to flow from ontology models into **production systems**, a new challenge emerges:

"How do we act on this knowledge?"

Executable Knowledge answers this question.

It is the discipline of encoding semantic knowledge with sufficient logical constraints to be interpreted deterministically and to drive action -- closing the loop between reasoning and execution.

Executable Knowledge transforms an ontology from a system that *knows* into a system that *acts*.

This chapter examines Exercise 22 through 26 from Michael DeBellis' *Protégé 5 New OWL Pizza Tutorial* through the **lens of executability**. While **Exercise 20** and **Exercise 21 (Chapter (23))** demonstrated the power of reasoning, **Exercise 22-26** introduce the OWL constructs that make reasoning *actionable*: universal restrictions, closure axioms, enumerated classes, `hasValue` restriction, and cardinality restrictions.

These constructs are the **execution primitives** of an ontology. They are what separate a descriptive model from an operational one.

Along the way, we will examine executability from multiple complementary perspectives: mathematical, engineering, architectural, and industrial -- revealing that executable knowledge is not a new invention, but the natural culmination of the entire SKDL arc.

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24.1 Chapter Introduction -- From Intelligence to Action

24.1.1 The Journey So Far

The first six stages of the Semantic Knowledge Development Lifecycle established the foundations of ontology engineering.

Stage 1 -- Conceptual Modeling identified the concepts that define a knowledge domain and organized them into a coherent semantic taxonomy.

Stage 2 -- Semantic Description enriched those concepts with formal meaning using Description Logic, OWL restrictions, and class expressions.

Stage 3 -- Knowledge Reuse transformed validated semantic knowledge into reusable assets that could be shared, extended, and specialized systematically.

Stage 4 -- Semantic Governance established the policies, standards, and processes required to maintain semantic integrity across an evolving knowledge ecosystem.

Stage 5 -- Semantic Validation verified that the ontology is logically consistent, semantically complete, and fit for purpose before deployment.

Stage 6 -- Semantic Reasoning derived new knowledge through automated inference -- computing the logical consequences of the asserted axioms.

By the end of **Stage 6**, the `Pizza` ontology had become intelligent. The reasoner auto-classified `MargheritaPizza` and other cheese-topped pizzas as subclasses of `CheesyPizza`. The inferred hierarchy now contained knowledge that was never explicitly asserted.

The ontology had become **intelligent**.

But intelligent is **not action**.

An intelligent ontology that cannot act is like a brilliant advisor who never speaks. It contains valuable knowledge, but it does not use that knowledge to change the world.

It is passive, not active!

This is where **Stage 7 -- Executable Knowledge** enters the lifecycle.

24.1.2 The New Question

Reasoning asked: ***What else can this model tell us?***

Executable Knowledge asks: ***What should happen, given what we know?***

The distinction is fundamental.

Reasoning derives new knowledge from a validated ontology. Executable Knowledge uses that derived knowledge to **drive action**.

Without reasoning, execution has no foundation.

Without execution, reasoning has no consequence.

Together, they produce actionable intelligence.

24.1.3 The Central Insight

The central insight of this chapter is deceptively simple:

Every OWL axiom you write is a potential execution rule. The question is not whether your ontology can be executed -- it is whether you have written the axioms that make execution possible.

The reasoner infers facts.

But inference alone does not trigger action.

Action requires:

- **Determinism** -- the ontology must produce a single, unambiguous result
- **Completeness** -- the ontology must guarantee that required data is present
- **Closure** -- the ontology must define boundaries, not leave them open

These are not automatic. They must be **engineered** through deliberate axioms.

Exercise 22-26 introduce these axioms.

24.2 Learning Objectives

After completing this chapter, you should be able to achieve the following learning outcomes:

Knowledge:

- Explain why Executable Knowledge is the seventh stage of the Semantic Knowledge Development Lifecycle (SKDL)
- Distinguish between descriptive knowledge and executable knowledge
- Understand the role of universal restrictions, closure axioms, enumerated classes, `hasValue` restrictions, and cardinality restrictions in enabling execution
- Explain how the Open World Assumption affects executability
- Understand how OWL constructs become execution primitives in the EKA framework

Practical Skills:

- Create universal restrictions (`only`) in Protégé
- Define closure axioms using combinations of `only` and cardinality restrictions
- Create enumerated classes using `EquivalentTo`
- Define `hasValue` restrictions to bind properties to specific individuals
- Define cardinality restrictions (`min` , `max` , `exactly`) to enforce data completeness
- Test executability by running the reasoner against test individuals

Mathematical Perspective:

- Understand how closure axioms convert open-world semantics to closed-world specifications
- Explain the mathematical relationships between `some` , `only` , `exactly` , and `value` restrictions
- Understand how enumeration defines a finite value domain
- Explain how cardinality restrictions constrain the extension of a class

Historical Perspective:

- Trace the concept of executability from early expert systems to modern rule engines
- Understand how the "knowledge is action" principle connects to cybernetics, control theory, and business process management

Engineering Perspective:

- Understand execution as the operationalization of semantic knowledge
- Explain why closure axioms are necessary for deterministic execution
- Map executable knowledge to the Θ (Trigger) and Φ (Execution) layers in the EKA framework
- Apply executable knowledge to industrial scenarios (manufacturing, compliance, safety)

24.3 Positioning Within SKDL -- Stage 7 Highlighted

As introduced in **Chapter (17)**, ontology engineering progresses through seven stages of the Semantic Knowledge Development Lifecycle (SKDL), each stage contributing a distinct engineering objective toward the construction of **executable semantic knowledge systems**.

Having completed **Stage 6 -- Semantic Reasoning** in the previous chapter, this chapter advances to **Stage 7 -- Executable Knowledge**.

The objective of this stage is to operationalize semantic knowledge -- to encode the ontology with the constraints required for **deterministic, actionable** execution.

At the completion of **Stage 6**, the `Pizza` ontology contained an inferred hierarchy that included knowledge derived by the reasoner. However, the ontology was still **descriptive**. It could tell us that `MargheritaPizza` is a subclass of `CheesyPizza` . But it could not yet tell us what to **do** about it.

Stage 7 addresses this gap by introducing:

- **Universal Restrictions** -- encoding exclusion and deterministic classification
- **Closure Axioms** -- converting open-world descriptions to closed-world specifications
- **Enumerated Classes** -- defining finite value domains
- **hasValue Restrictions** -- binding properties to specific values
- **Cardinality Restrictions** -- enforcing data completeness

Rather than introducing new semantic constructs, this stage establishes the engineering discipline required to make semantic knowledge actionable.

The primary deliverable of **Stage 7** is therefore:

not a new logical axiom, but an executable ontology -- one whose inferred knowledge can drive triggers, execute actions, and produce real-world effects.

Within the EKA framework, **Stage 7** establishes the Θ -- **Trigger** and Φ -- **Execution** layers by providing the semantic conditions that fire triggers and the action bindings that execute them.

The progression across all seven stages can now be seen in full:

SKDL Stage	Primary Outcome	EKA Contribution
Stage 1 -- Conceptual Modeling	Identify concepts and organize them into a coherent semantic taxonomy	K (Knowledge Graph) (Structure)
Stage 2 -- Semantic Description	Enrich concepts with formal semantic meaning through Description Logic and OWL restrictions	K (Meaning) + R (Readiness)
Stage 3 -- Knowledge Reuse	Transform validated semantic knowledge into reusable assets	K (Composable) + Γ (Prepared)
Stage 4 -- Semantic Governance	Maintain semantic integrity through policies, standards, and lifecycle management	Γ (Governance)
Stage 5 -- Semantic Validation	Verify correctness before deployment through consistency checking and constraint validation	Γ + K (Validation)
Stage 6 -- Semantic Reasoning	Derive new knowledge through automated inference	R (Reasoning & Rules)
Stage 7 -- Executable Knowledge	Operationalize semantic knowledge for deterministic action	Θ (Trigger) + Φ (Execution)

Each stage has built upon the previous one.

- **Stage 1** gave us concepts.
- **Stage 2** gave us meaning.
- **Stage 3** gave us reusability.
- **Stage 4** gave us governance.
- **Stage 5** gave us trust.
- **Stage 6** gave us intelligence.
- **Stage 7** now gives us **action**.

24.4 What is Executable Knowledge?

24.4.1 A Working Definition

Executable Knowledge is semantic knowledge that has been encoded with sufficient logical constraints to be interpreted deterministically and to drive action.

It answers the question:

"What should happen, given what we know?"

Not "what is true?" (that is reasoning).

Not "what is correct?" (that is validation).

But "what should be **done**?"

This is fundamentally an **operationalization** activity. It is the moment when the ontology becomes a decision system -- when its axioms do not merely describes the world but specify what to do about it.

Key characteristics of executable knowledge:

- **Deterministic** -- Given the same facts, the same action is triggered. No ambiguity.
- **Closed** -- The ontology specify boundaries, not open-ended descriptions.
- **Complete** -- The ontology guarantees that required data is present before acting.
- **Actionable** -- The ontology binds semantic conditions to executable actions.

Key Insight:

"Reasoning tells us what is true. Executable Knowledge tells us what to do about it. Without executability, the ontology is a library. With it, the ontology is a decision system."

24.4.2 The Three Layers of Semantic Knowledge

Semantic knowledge is an ontology operates at three distinct layers:

Layer	Question	OWL Constructs	Stage
Description	What is true?	SubClassOf , some , only	Stage 1-2
Validation	What is trustworthy?	Disjointness, domain/range, SHACL	Stage 4-5
Execution	What should happen?	Closure axioms, cardinality, `hasValue, enumerated classes	Stage 7

The **description layer** defines concepts. The **validation layer** ensures correctness. The **execution layer** enables action.

Without the execution layer, the ontology is correct but passive. With it, the ontology is actionable.

Key Insight:

"Description without validation is unchecked. Validation without execution is theoretical. Execution without description and validation is dangerous. The three layers together form a complete semantic system."

24.4.3 Executable Knowledge vs. Reasoning vs. Validation

It is important to distinguish executability from reasoning and validation:

Activity	Question	Timing	Output
Validation	"Is this model correct?"	Before reasoning	Pass/fail -- the ontology is consistent or not
Reasoning	"What else can this model tell us?"	After validation	Inferred facts -- new classification, relationships

Activity	Question	Timing	Output
Executable Knowledge	"What should happen?"	After reasoning	Actions -- triggered operations, workflow steps

Validation ensures correctness. **Reasoning** derives new knowledge. **Executable Knowledge** acts on that knowledge.

The key distinction is this: **reasoning produces facts; executability produces actions.**

A reasoned ontology knows that a patient is `HighRiskPatient`, then an executable ontology sends an alert to the clinician.

Key Insight:

"Reasoning is the brain. Execution is the hand. A brain without a hand is a philosopher. A hand without a brain is a hazard. Together, they produce intelligent action."

24.4.3 The Open World Assumption and Executability

The Open World Assumption (OWA) is foundational to OWL. It states that just because something is not asserted does not mean it is false. This is correct for reasoning -- but it creates a problem for execution.

Consider a pizza with `hasTopping some MozzarellaTopping`. Under OWA, the reasoner cannot conclude that the pizza has **only** mozzarella. There might be other toppings we simply don't know about.

For reasoning, this is correct.

For execution, it is problematic.

An execution system that recommends a pizza based on an incomplete description may recommend the wrong thing.

Key Insight:

"The Open World Assumption is correct for reasoning and dangerous for execution. Execution requires the closed world. **Closure axioms** are the bridge."

This tension is the reason ****Exercise 22-26** exist. They introduce the OWL constructs that allow us to **close** the world when execution requires it.

24.5 Exercise 22 -- Universal Restrictions and Deterministic Classification

Having established the principles of executable knowledge, we now turn to its practical implementation through **Exercise 22** in Michael DeBellis' *Protégé 5 New OWL Pizza Tutorial*.

This exercise represents the first hands-on activity of **Stage 7 -- Executable Knowledge** within the Semantic Knowledge Development Lifecycle (SKDL).

Unlike previous exercises, which introduced individual OWL constructs or semantic modeling patterns, **Exercise 22** focuses on the construct that makes deterministic classification possible: **the universal restriction**.

24.5.1 The Engineering Objective

Exercise 22 introduces that **universal restriction**, represented in OWL Manchester Syntax by the keyword `only`.

The universal restriction specifies that **all** values of a property must satisfy a specified condition. It does not require that any values exist -- only that, if they exist, they must satisfy the condition.

The definition for a vegetarian pizza, for example, might be:

```
Class: VegetarianPizza
  SubClassOf: Pizza
  SubClassOf: hasTopping only (CheeseTopping or VegetableTopping)
```

This means: "Every topping on a vegetarian pizza must be either a cheese topping or a vegetable topping."

The engineering objective of **Exercise 22** is to introduce the `only` construct and demonstrate how it enables deterministic classification -- the ability to **execute** as well as include.

Key Insight:

- Existential restrictions tell the reasoner what must be present.
- Universal restrictions tell the reasoner what must not be present.

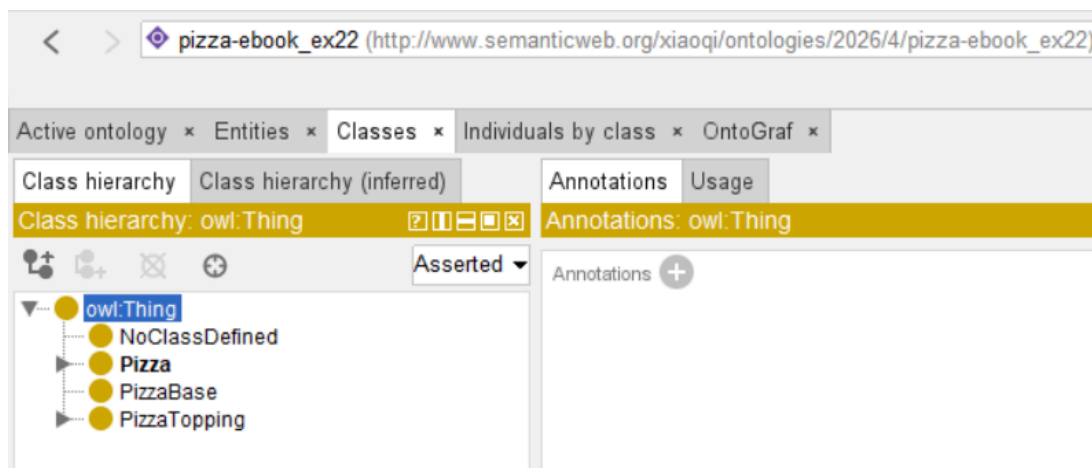
Execution needs both.

24.5.2 Implementation: Creating the Universal Restriction

Let's walk through the implementation step by step. As you'll see, the exercise **deliberately** includes a moment where the expected result does **not** appear -- and that moment is the most important part of the lesson.

Step 1: Open the `Pizza` ontology

The RDF file for this exercise is named as `pizza-ebook_ex22.rdf`. You may find it in the ebook's `\rdf` folder. Open it in Protégé. Navigate to the Classes tab. (Tip: you may open `pizza-ebook_ex21.rdf` as the starting ontology snapshot.)



Step 2: Select `VegetarianPizza`

Select `VegetarianPizza` in the class hierarchy. In a previous exercise, we have already defined this class as

```
Class: VegetarianPizza
  SubClassOf: Pizza
  SubClassOf: hasBase some PizzaBase
  EquivalentTo: hasTopping only VegetableTopping
```

We will update it now to match **Exercise 22**.

The screenshot shows the Protege interface with the 'pizza-ebook_ex22' ontology loaded. The 'Classes' tab is active, and the 'Class hierarchy' view is selected. The hierarchy shows 'VegetarianPizza' as a subclass of 'Pizza'. The 'Annotations' tab is also visible, showing the 'rdfs:label' and 'rdfs:comment' for 'VegetarianPizza'. The 'Description' tab is selected, showing the 'SubClass Of' section with 'Pizza' as a parent class.

Step 3: Add the universal restriction

1. In the Description view, locate the **SubClassOf** section
2. Click the Add icon (+)
3. In the Class Expression Editor, enter `hasTopping only (CheeseTopping or VegetableTopping)`
4. Click OK

After this step, `VegetarianPizza` has:

```
Class: VegetarianPizza
SubClassOf: Pizza
SubClassOf: hasTopping only (CheeseTopping or VegetableTopping)
```

The ontology after this step is as below:

The screenshot shows the Protege interface after adding the universal restriction. The 'Classes' tab is active, and the 'Class hierarchy' view is selected. The hierarchy shows 'VegetarianPizza' as a subclass of 'Pizza'. The 'Annotations' tab is also visible, showing the 'rdfs:label' and 'rdfs:comment' for 'VegetarianPizza'. The 'Description' tab is selected, showing the 'SubClass Of' section with 'Pizza' and 'hasTopping only (CheeseTopping or VegetableTopping)' as parent classes. The new restriction is highlighted with a red box.

Step 4: Add a test individual

To observe the effect of the universal restriction, let's add a test individual.

1. Navigate to the `Individual by class` tab

Active ontology x Entities x Classes x **Individuals by class** x OntoGraf x

Class hierarchy: Vegetar [?] [I] [E] [C] [X]

Annotations Usage

Annotations:

Annotations +

Description: [?] [I] [E] [C] [X] Property assertions:

Types +

Same Individual As +

Different Individuals +

Object property assertions +

Data property assertions +

Negative object property assertions +

Negative data property assertions +

Direct instances: [?] [I] [E] [C] [X]

For: ● VegetarianPizza

◆ VegetarianPizza1

2. Keep class `Pizza` selected in the class hierarchy, create a new individual named `TestVegetarianPizza`

Active ontology x Entities x Classes x **Individuals by class** x OntoGraf x

Class hierarchy: Pizza [?] [I] [E] [C] [X]

Annotations Usage

Annotations:

Annotations +

Description: [?] [I] [E] [C] [X] Property assertions:

Types +

Same Individual As +

Object property assertions +

Data property assertions +

Direct instances: [?] [I] [E] [C] [X]

For: ● Pizza

◆ Pepperoni

Create a new Named individual

Name TestVegetarianPizza

IRI manticweb.org/xiaoqi/ontologies/2026/4/pizza-ebook_ex22#TestVegetarianPizza

New entity options...

OK Cancel

3. After click `ok` , notice the type of individual is `Pizza` now

The screenshot shows the OntoGraf interface with the following components:

- Active ontology**: Pizza
- Entities**: Classes
- Individuals by class**: TestVegetarianPizza
- Annotations**: TestVegetarianPizza
- Class hierarchy**: Pizza (selected), NoClassDefined, CheesyPizza, NamedPizza, NonVegetarianPizza, VegetarianPizza, PizzaBase, PizzaTopping.
- Annotations**: TestVegetarianPizza (empty list).
- Description**: TestVegetarianPizza
- Property**: Object prop, Data prop, Negative ot, Negative ds.
- Types**: Pizza (selected).
- Same Individual As**: +
- Different Individuals**: +
- Direct instances**: TestVegetarianPizza (selected), Pepperoni.

4. Add the property assertion `hasTopping some MozzarellaTopping`

The screenshot shows the OntoGraf interface with the following components:

- Description**: TestVegetarianPizza
- Property assertions**: TestVegetarianPizza
- Types**: Pizza (selected).
- Object property assertions**: +
- Data property assertions**: +
- Object restriction creator**: TestVegetarianPizza
- Class expression editor**: `hasTopping some MozzarellaTopping`

5. Add the property assertion `hasTopping some TomatoTopping`

The screenshot shows the OntoGraf interface with the following components:

- Description**: TestVegetarianPizza
- Property assertions**: TestVegetarianPizza
- Types**: `hasTopping some MozzarellaTopping` (selected), Pizza.
- Object property assertions**: +
- Data property assertions**: +
- Object restriction creator**: TestVegetarianPizza
- Class expression editor**: `hasTopping some TomatoTopping`

6. Now ensure your ontology is looked like this

Active ontology × Entities × Classes × Individuals by class × OntoGraf ×

Class hierarchy: VegetarianPizza ? [] [] [] []

Annotations Usage

Annotations: TestVegetarianPizza

Annotations +

Description: TestVegetarianPizza ? [] [] [] []

Types +

- hasTopping some MozzarellaTopping ? @ × ○
- hasTopping some TomatoTopping ? @ × ○
- Pizza ? @ × ○

Same Individual As +

Direct instances: TestVegetarianPizza ? [] [] [] []

For: ● VegetarianPizza

◆ VegetarianPizza1

Step 5: Synchronize the reasoner

Select Reasoner > Synchronize reasoner (Ctrl+R).

Observe the result.

Active ontology × Entities × Classes × Individuals by class × OntoGraf ×

Class hierarchy: VegetarianPizza ? [] [] [] []

Annotations Usage

Annotations: TestVegetarianPizza ? [] [] [] []

Annotations +

Description: TestVegetarianPizza ? [] [] [] []

Types +

- hasTopping some MozzarellaTopping ? @ × ○
- hasTopping some TomatoTopping ? @ × ○
- Pizza ? @ × ○
- CheesyPizza ? @

Same Individual As +

Direct instances: TestVegetarianPizza ? [] [] [] []

For: ● VegetarianPizza

◆ VegetarianPizza1

Property assertions

Object property as

Data property as

Negative object pr

Negative data prop

With ontology designed so far, the reasoner classifies `TestVegetarianPizza` as a `CheesyPizza`, because we have defined `CheesyPizza` earlier using `EquivalentTo` (defined class) and `TestVegetarianPizza` has toppings `MozzarellaTopping` (a `CheeseTopping`):

```
Class: CheesyPizza
  Equivalent To: Pizza and (hasTopping some CheeseTopping)
  SubClass Of: Pizza
```

Class hierarchy: CheesyPizza

- owl:Thing
 - NoClassDefined
 - Pizza
 - CheesyPizza
 - NamedPizza
 - NonVegetarianPizza
 - VegetarianPizza
 - PizzaBase
 - PizzaTopping

Annotations: CheesyPizza

Annotations +

Description: CheesyPizza

Equivalent To +

Pizza and (hasTopping some CheeseTopping)

SubClass Of +

Pizza

But till now the `VegetarianPizza` is still under only `SubClassOf` (primitive class), so base on Open World Assumption (OWA), the reasoner cannot infer `TestVegetarianPizza` as a type of `VegetarianPizza`.

Step 6: Convert `VegetarianPizza` to a defined class

This is the crucial step. Select `VegetarianPizza` in the Classes tab, then select `Edit > Convert to defined class` (or from right-click menu).

Active ontology * Entities * Classes * Individuals by class * OntoGraf *

Class hierarchy Class hierarchy (inferred) Annotations Usage

Class hierarchy: VegetarianPizza

- owl:Thing
 - NoClassDefined
 - Pizza
 - CheesyPizza
 - NamedPizza
 - NonVegetarianPizza
 - VegetarianPizza
 - PizzaBase
 - PizzaTopping

Annotations: VegetarianPizza

Annotations +

rdfs:label Vegetarian Pizza

rdfs:comment A pizza containing only non-meat toppings, suitable for vegetarian diets.

Context menu:

- Add subclass... Ctrl-E
- Add sibling class... Ctrl+Shift-E
- Add subclasses...
- Duplicate class... Ctrl+Shift-C
- Change IRI (Rename)... Ctrl-U
- Copy Ctrl-C
- Copy IRI
- Copy OBO Id
- Copy display name
- Copy as Markdown
- Copy sub-hierarchy as tab indented text Ctrl+Alt-C
- Show IRI in Web browser
- Make primitive siblings disjoint Ctrl-J
- Convert to primitive class Ctrl-P
- Convert to defined class Ctrl-D
- Remove disjoint classes ax Converts the superclasses of the selected class to an equivalent class.

Observe what happens in the Description view. The `SubClassOf` axiom for the universal restriction is moved to `EquivalentTo`:

Class: VegetarianPizza
 Equivalent To: Pizza and (hasTopping only (CheeseTopping or VegetableTopping))
 SubClass Of: Pizza

The screenshot shows the Protege interface with the 'Classes' tab selected. The 'Class hierarchy' panel on the left shows a tree structure where 'VegetarianPizza' is highlighted. The 'Annotations' panel on the right shows the following information for 'VegetarianPizza':

- Annotations:**
 - rdfs:label:** Vegetarian Pizza
 - rdfs:comment:** A pizza containing only non-meat toppings, suitable for vegetarian diets.
- Description:** VegetarianPizza
- Equivalent To:** Pizza and (hasTopping only (CheeseTopping or VegetableTopping))
- SubClass Of:** Pizza

Note

The `vegetarianPizza` is now shown with three horizontal stripes in the class icon -- the visual indicator that it is a defined class.

Step 7: Synchronize the reasoner again

Select Reasoner > Synchronize reasoner (Ctrl+R)

Step 8 -- Observe the result -- and be surprised!

Look at `TestVegetarianPizza` in the **Individual by class** tab.

The screenshot shows the Protege interface with the 'Individuals by class' tab selected. The 'Class hierarchy' panel on the left shows a tree structure where 'Pizza' is highlighted. The 'Direct instances' panel on the right shows the following information for 'TestVegetarianPizza':

- Annotations:** TestVegetarianPizza
- Description:** TestVegetarianPizza
- Types:**
 - hasTopping some MozzarellaTopping
 - hasTopping some TomatoTopping
 - Pizza
 - CheesyPizza

You will see that the reasoner has kept classifying `TestVegetarianPizza` as:

-  `CheesyPizza` (because it has at least one cheese topping -- mozzarella)

But it has **NOT** classified `TestVegetarianPizza` as:

-  `VegetarianPizza`

This is surprising. `TestVegetarianPizza` has only mozzarella and tomato toppings. Both are vegetarian. **Why is it not a `VegetarianPizza` ?**

Step 9 -- Pause and understand

This result is not a bug. It is still the **Open World Assumption (OWA)** in action, although we have converted `VegetarianPizza` from primitive class to defined class, and it is one of the most important lessons in ontology engineering.

When we define `VegetarianPizza` as:

```
Pizza and (hasTopping only (CheeseTopping or VegetableTopping))
```

we are saying: **"All toppings must be `CheeseTopping` or `VegetableTopping` .**

But `TestVegetarianPizza` only has two *asserted* toppings. Under the Open World Assumption, the reasoner cannot conclude that these are the *only* toppings -- there might be other toppings that exist in the world (somewhere, anywhere) not been added to the ontology yet.

The reasoner can only classify an individual into a defined class when it has enough information to prove that the individual satisfies the definition. And it does not have enough information here.

Why `CheesyPizza` works but `VegetarianPizza` doesn't:

Class	Definition	Reasoner behavior
CheesyPizza	Pizza and (hasTopping some CheeseTopping)	Existential restriction. The reasoner only needs to find one cheese topping -- which exists (<code>MozzarellaTopping</code>).  Classified.
VegetarianPizza	Pizza and (hasTopping only (CheeseTopping or VegetableTopping))	Universal restriction. The reasoner must verify that all toppings satisfy the constraint -- but OWA prevents it from knowing that no other toppings exist.  Not classified.

Something unexpected has happened. `TestVegetarianPizza` has only mozzarella topping (a `CheeseTopping`) and tomato topping (a `VegetableTopping`). We defined `VegetarianPizza` as a defined class using a universal restriction. And yet the reasoner has classified `TestVegetarianPizza` as a `CheesyPiza` but **not** as a `VegetarianPizza` .

This is not a bug in Protégé, and it is even not an error in our ontology. It is a fundamental property of OWL that every ontology engineer must understand. The reason for this behavior -- and the construct that resolves it -- is the subject of the next exercise.

24.5.3 The Result: What the Reasoner Does Not Do

When the reasoner is synchronized, something important -- or rather, something important does **not** happen.

What to observe: (let's recap)

- `CheesyPizza` is correctly inferred for `TestVegetarianPizza` (highlighted in yellow, indicating a reasoner inference).
- `TestVegetarianPizza` is **not** classified as a `VegetarianPizza` .

The reasoner has all the information it needs to conclude that `TestVegetarianPizza` is a `VegetarianPizza` -- it has mozzarella (a `CheeseTopping`) and tomato (a `VegetableTopping`), both of which are permitted by the universal restriction on `VegetarianPizza` . **And yet it does not make that inference.**

This is unexpected. We defined `VegetarianPizza` as a defined class with the universal restriction `hasTopping only (CheeseTopping or VegetableTopping)` . We created `TestVegetarianPizza` with exactly the kinds of toppings that restriction permits. Intuitively, from human being's perspective, the reasoner should classify it as a `VegatarianPizza` .

But it does not.

Key Insight:

"The universal restriction tells the reasoner what is allowed. It does not tell the reasoner what is *exhaustively* present. This distinction is the difference between a description and a specification.

Why this happens -- and what construct resolves it -- is the subject of **Exercise 23**.

24.5.4 Why the Universal Restriction Is Not Enough

To understand why the reasoner does not classify `TestVegetarianPizza` as a `VegetarianPizza` , we need to examine how universal restrictions behave under the **Open World Assumption (OWA)**.

The universal restriction:

```
VegetarianPizza
  EquivalentTo
    Pizza and (hasTopping only (CheeseTopping or VegetableTopping))
```

means: "If something is a `VegetarianPizza` , then every one of its toppings must be either a `CheeseTopping` or a `VegetableTopping` ."

For the reasoner to conclude the `TestVegetarianPizza` **is** a `VegetarianPizza` , it must verify that the universal restriction holds -- that *all* of `TestVegetarianPizza` 's toppings are cheese or vegetable. But under the Open World Assumption, the reasoner cannot assume that the two toppings it knows about are the *only* toppings. There might be other toppings that exist but have not been asserted.

So the reasoner cannot prove that `TestVegetarianPizza` satisfies the universal restriction, and therefore cannot classify it as a `VegetarianPizza` .

This is not a flaw in the ontology. It is a faithful consequence of how OWL is designed. The Open World Assumption is what makes OWL suitable for open, federated domains -- but it is also what makes universal restriction insufficient for classification on their own.

Key Insight:

"The universal restriction tells the reasoner: 'If you find a `VegetarianPizza` , check that all its toppings are either cheese or vegetable.' It does not tell the reasoner: 'If you fine a pizza whose known toppings are all cheese or vegetable, it is a `VegetarianPizza` .' The second statement requires more information than the universal restriction alone can supply."

24.5.5 Engineering Perspective: Exercise 22 as a Case-Study

From the perspective of Protégé, **Exercise 22** involves adding a single restriction and creating a test individual. (Note: the adding test individual is the addition from tutorial's exercise to trigger more discussion.) It appears to be a small modeling activity.

From the perspective of ontology engineering, however, something significant has occurred. The exercise has introduced the first executive primitive -- the universal restriction -- *and* surfaced the first significant engineering obstacle:

the OWA between what a universal restriction describes and what execution requires.

The engineering contribution of **Exercise 22** can be summarized as follows:

Engineering Perspective	Contribution from Exercise 22
SKDL Stage	Stage 7 -- Executable Knowledge
Primary Objective	Introduce the universal restriction as an execution primitive
Modeling Activity	Add <code>hasTopping only (VegetableTopping or CheeseTopping)</code> to <code>VegetarianPizza</code> ; convert to defined class; add <code>TestVegetarianPizza</code> individual
Engineering Outcome	Demonstrate the universal restriction's expressive power -- and its limitation under OWA

Engineering Perspective	Contribution from Exercise 22
Preparation for Later Sections	Set up the OWL problem that Exercise 23 will resolve through closure axioms

Key Insight:

"Exercise 22 is a lesson in what the universal restriction can and cannot do. It can **exclude**. It cannot, on its own, **proved** exhaustiveness. The task belongs to the closure axiom."

24.6 Exercise 23 -- Closure Axioms and the Open World Assumption

In **Exercise 22**, we saw a surprising result: even though `VegetarianPizza` was correctly defined with a universal restriction, the reasoner did not classify `TestVegetarianPizza` as a `VegetarianPizza`. The reason was the Open World Assumption (OWA) -- the reasoner could not conclude that `TestVegetarianPizza` has *only* mozzarella and tomato toppings, because it could not rule out the existence of other toppings that had not been asserted.

This is the same phenomenon that Michael DeBellis demonstrates in Section 4.13 of the `pizza` Tutorial, where he shows that `MargheritaPizza` and `SohoPizza` -- despite having only cheese and vegetable toppings -- are not automatically classified as subclasses of `VegetarianPizza` by the reasoner.

The construct that resolves this problem is called the **closure axiom**.

This exercise follows the tutorial's **Exercise 23**, and then extends it slightly to tie back to the `TestVegetarianPizza` individual from the previous exercise.

24.6.1 The Engineering Objective

Exercise 23 introduces the **closure axiom** -- a universal restriction that explicitly specifies the complete set of values a property may take for a class or an individual.

A closure axiom for `MargheritaPizza` would be:

```
Class: MargheritaPizza
  SubClassOf: hasTopping only (MozzarellaTopping or TomatoTopping)
```

This says: "Every toppings on a `MargheritaPizza` must be either `MozzarellaTopping` or `TomatoTopping`, and no other toppings exist."

By adding this axiom, we close the world for `MargheritaPizza`. The reasoner now knows that the two toppings we asserted for `MargheritaPizza` are the *only* toppings it has. This is the information it needs to classify `MargheritaPizza` as a subclass of `VegetarianPizza`.

Key Insight:

"A closure axiom is the OWL equivalent of saying 'this is the complete picture.'"

- Without closure, the reasoner remains cautious.
- With closure, the reasoner can be decisive."

24.6.2 Implementation: Adding Closure Axioms to MargheritaPizza and SohoPizza

We begin with the tutorial's **Exercise 23**, including the discussion in tutorial's Section 4.13 (before the exercise), which applies closure axioms at the class level.

Step 1: Open the Pizza ontology

Open `pizza-ebook_ex22.rdf` in Protégé -- the file from the previous exercise, which already contains `TestVegetarianPizza` and the inferred-but-incomplete classification we observed.

The screenshot shows the Protege application window with the 'pizza-ebook_ex22' ontology loaded. The 'Ontology header' tab is active, displaying the following information:

- Ontology IRI:** http://www.semanticweb.org/xiaoqi/ontologies/2026/4/pizza-ebook_ex22
- Ontology Version IRI:** e.g. http://www.semanticweb.org/xiaoqi/ontologies/2026/4/pizza-ebook_ex22/1.0.0
- Annotations:** A list of annotations, including `dcterms:license` with the value `https://creativecommons.org/licenses/by-sa/4.0/`.

The 'Ontology metrics' tab is also visible, showing the following data:

Metrics	
Axiom	136
Logical axiom count	76
Declaration axioms count	51
Class count	30
Object property count	4
Data property count	0
Individual count	16
Annotation Property count	3

The 'Class axioms' tab is also visible, showing the following data:

Class axioms	
SubClassOf	39
EquivalentClasses	2
DisjointClasses	4
GCI count	0
Hidden GCI Count	0

Note

The finished state of **Exercise 23** will be saved as `pizza-ebook_ex23.rdf`.

Step 2: Select `MargheritaPizza`

In the Classes tab, select `MargheritaPizza` in the class hierarchy.

The screenshot shows the Protege application window with the 'Classes' tab active. The 'Class hierarchy' is displayed on the left, and the 'Annotations' and 'Usage' tabs are active on the right.

Class hierarchy: The hierarchy shows the following classes:

- owl:Thing
- NoClassDefined
- Pizza
 - CheesyPizza
 - NamedPizza
 - AmericanaPizza
 - MargheritaPizza** (selected)
 - NonVegetarianPizza
 - VegetarianPizza
- PizzaBase
- PizzaTopping

Annotations: The annotations for `MargheritaPizza` are:

- `rdfs:comment`: A pizza that only has Mozzarella and Tomato toppings

Description: The description for `MargheritaPizza` is:

- Equivalent To: +
- SubClass Of: +
 - `hasTopping some MozzarellaTopping`
 - `hasTopping some TomatoTopping`
 - `NamedPizza`

So far, the `MargheritaPizza` is defined as:

Class: MargheritaPizza
 SubClassOf: NamedPizza
 SubClassOf: hasTopping some MozzarellaTopping
 SubClassOf: hasTopping some TomatoTopping

Step 3: Add the closure axiom to MargheritaPizza

1. In the Description view, locate the **SubClass Of** section

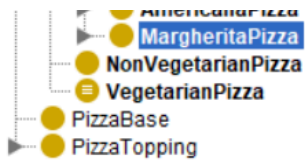
The screenshot shows the OntoGraf interface with the 'Class hierarchy' tab selected. The class hierarchy for MargheritaPizza is displayed on the left, showing its inheritance from NamedPizza and its relationship to PizzaBase and PizzaTopping. The 'SubClass Of' section is highlighted with a red box, showing the following axioms:

- hasTopping some MozzarellaTopping
- hasTopping some TomatoTopping
- NamedPizza

2. Click the Add icon (+)

The screenshot shows the OntoGraf interface with the 'Class hierarchy' tab selected. The class hierarchy for MargheritaPizza is displayed on the left. The 'SubClass Of' section is highlighted with a red box, and the 'Add' icon (+) is also highlighted. The 'Class expression editor' is visible at the bottom, showing the class expression 'MargheritaPizza'.

3. In the Class Expression Editor, enter: hasTopping only (MozzarellaTopping or TomatoTopping)



Description: MargheritaPizza

Equivalent To +

SubClass Of +

Class expression editor Class hierarchy Data restriction creator Object

hasTopping only (MozzarellaTopping or TomatoTopping)

4. Click ok .

After this step, MargheritaPizza has the following description:

Class: MargheritaPizza
 SubClassOf: NamedPizza
 SubClassOf: hasTopping some MozzarellaTopping
 SubClassOf: hasTopping some TomatoTopping
 SubClassOf: hasTopping only (MozzarellaTopping or TomatoTopping)

Class hierarchy Class hierarchy (inferred) Annotations Usage

Class hierarchy: MargheritaPizza

Annotations: MargheritaPizza

Annotations +

rdfs:comment
A pizza that only has Mozzarella and Tomato toppings

Description: MargheritaPizza

Equivalent To +

SubClass Of +

- hasTopping only (MozzarellaTopping or TomatoTopping)
- hasTopping some MozzarellaTopping
- hasTopping some TomatoTopping
- NamedPizza

Step 4: Add the closure axiom to SohoPizza

Repeat Step 2-3, but this time select SohoPizza and enter the axiom:

hasTopping only (MozzarellaTopping or TomatoTopping or ParmesanTopping or OliveTopping)

The screenshot shows the Protégé interface with the 'Class hierarchy' tab selected. The class hierarchy on the left shows 'SohoPizza' as a subclass of 'MargheritaPizza'. The 'Annotations' tab on the right shows the following information for 'SohoPizza':

- Annotations:** SohoPizza
- Annotations:** +
- rdfs:comment:** A pizza that has Mozzarella, Tomato, Olive, and Parmesan toppings. Created by cloning from MargheritaPizza. Additional restrictions: hasTopping some OliveTopping, hasTopping some ParmesanTopping. Source: MargheritaPizza.
- Description:** SohoPizza
- Equivalent To:** +
- SubClass Of:** +
 - hasTopping some MozzarellaTopping
 - hasTopping some OliveTopping
 - hasTopping some ParmesanTopping
 - hasTopping some TomatoTopping
 - MargheritaPizza

A small window titled 'SohoPizza' is open, showing the 'Class expression editor' with the following expression: `hasTopping only (MozzarellaTopping or TomatoTopping or ParmesanTopping or OliveTopping)`.

After this step, SohoPizza has the closure axiom that enumerates its full set of allowed toppings.

The screenshot shows the Protégé interface with the 'Class hierarchy' tab selected. The class hierarchy on the left shows 'SohoPizza' as a subclass of 'MargheritaPizza'. The 'Annotations' tab on the right shows the following information for 'SohoPizza':

- Annotations:** SohoPizza
- Annotations:** +
- rdfs:comment:** A pizza that has Mozzarella, Tomato, Olive, and Parmesan toppings. Created by cloning from MargheritaPizza. Additional restrictions: hasTopping some OliveTopping, hasTopping some ParmesanTopping. Source: MargheritaPizza.
- Description:** SohoPizza
- Equivalent To:** +
- SubClass Of:** +
 - hasTopping only (MozzarellaTopping or OliveTopping or ParmesanTopping or TomatoTopping)
 - hasTopping some MozzarellaTopping
 - hasTopping some OliveTopping
 - hasTopping some ParmesanTopping
 - hasTopping some TomatoTopping
 - MargheritaPizza

The 'hasTopping only' restriction is highlighted with a red box.

Step 5: Synchronize the reasoner

Select Reasoner > Synchronize reasoner (Ctrl+R)

Step 6: Observe the result

Look at the **Class hierarchy (inferred)** tab.

The reasoner now classifies both `MargheritaPizza` and `SohoPizza` (subclass of `MargheritaPizza` still) as subclasses of `VegetarianPizza`. **This is the result we were looking for:**

The screenshot shows the OntoGraf interface with the 'Classes' tab selected. The 'Class hierarchy' panel on the left shows a tree structure where `VegetarianPizza` is a subclass of `Pizza`, and `MargheritaPizza` and `SohoPizza` are subclasses of `VegetarianPizza`. The 'Inferred' tab is selected, and a red box highlights the `VegetarianPizza` node. The right panel shows the 'Annotations' for `VegetarianPizza`, including `rdfs:label` 'Vegetarian Pizza' and `rdfs:comment` 'A pizza containing only non-meat toppings, suitable for vegetarian diets.' The 'Description' panel shows the logical definition: `Pizza and (hasTopping only (CheeseTopping or VegetableTopping))`.

Why does it work now? Because the closure axiom supplies the missing information (knowledge) -- it tells the reasoner that *only* the enumerated toppings exist for these classes. Without that information, the reasoner could not verify that the universal restriction on `VegetarianPizza` held for `MargheritaPizza` and `SohoPizza`.

Key Insight:

"The reasoner did not change. The definition of `VegetarianPizza` did not change. Only the **world** changed -- from **open** to **closed**. Once the world is closed, the reasoner can make the inference."

24.6.3 Extending the Demonstration: Closure at the Individual Level

We have now seen closure axioms applied at the **class** level, following the tutorial. But our `TestVegetarianPizza` individual from **Exercise 22** is still unclassified. Let's resolve that too, using the same principle.

Step 1: Select `TestVegetarianPizza` individual

Navigate to the **Individual by class** tab and select `TestVegetarianPizza`.

The screenshot shows the OntoGraf interface with the 'Individuals by class' tab selected. A red box highlights the 'Individuals by class' tab, and a red arrow points from it to the 'TestVegetarianPizza' instance in the 'Direct instances' list. The 'Class hierarchy' on the left shows 'Pizza' as a subclass of 'owl:Thing'. The 'Direct instances' list shows 'Pepperoni' and 'TestVegetarianPizza' as instances of 'Pizza'. The 'Description' view on the right shows the 'TestVegetarianPizza' class with its types: 'hasTopping some MozzarellaTopping', 'hasTopping some TomatoTopping', 'Pizza', and 'CheesyPizza'.

Step 2: Add the closure axiom

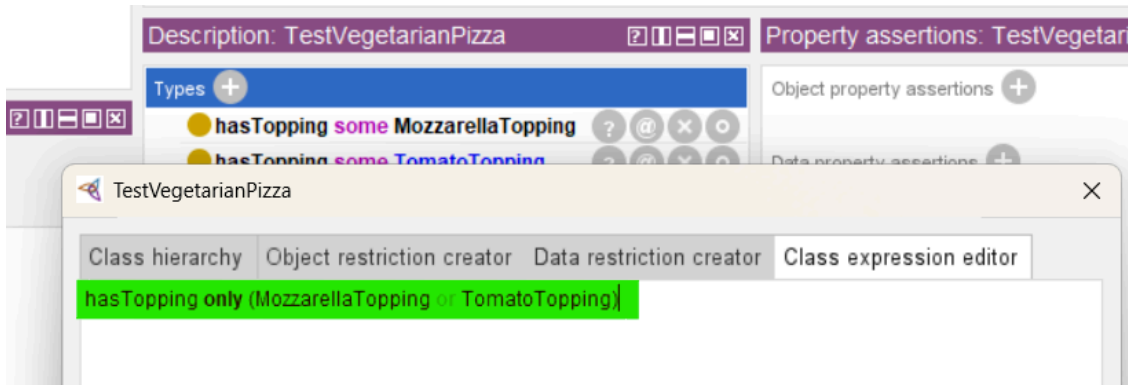
1. In the Description view, locate the **Types** section

This screenshot shows the 'Description' view of 'TestVegetarianPizza'. A red box highlights the 'Types' section, and a red arrow points from it to the 'TestVegetarianPizza' instance in the 'Direct instances' list. The 'Types' section lists: 'hasTopping some MozzarellaTopping', 'hasTopping some TomatoTopping', 'Pizza', and 'CheesyPizza'. The 'Direct instances' list shows 'Pepperoni' and 'TestVegetarianPizza' as instances of 'Pizza'.

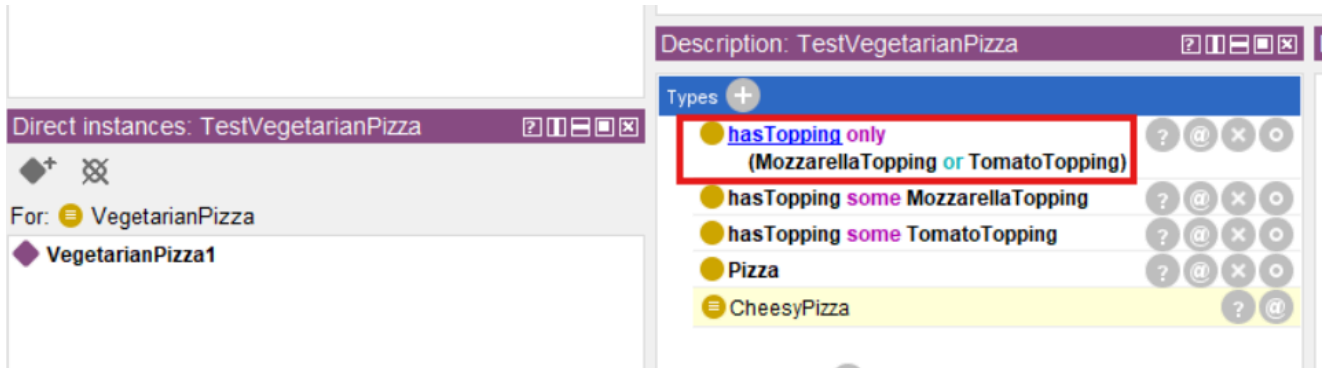
2. Click the Add icon (+)

This screenshot shows the 'Class expression editor' for 'TestVegetarianPizza'. A red box highlights the 'Add' icon (+) in the 'Types' section. The 'Class expression editor' is open, showing the 'Class hierarchy' tab. The 'Class expression editor' has tabs for 'Class hierarchy', 'Object restriction creator', 'Data restriction creator', and 'Class expression editor'. The 'Class expression editor' is currently active, and the 'TestVegetarianPizza' class is selected.

3. In the Class Expression Editor, enter: hasTopping only (MozzarellaTopping or TomatoTopping)



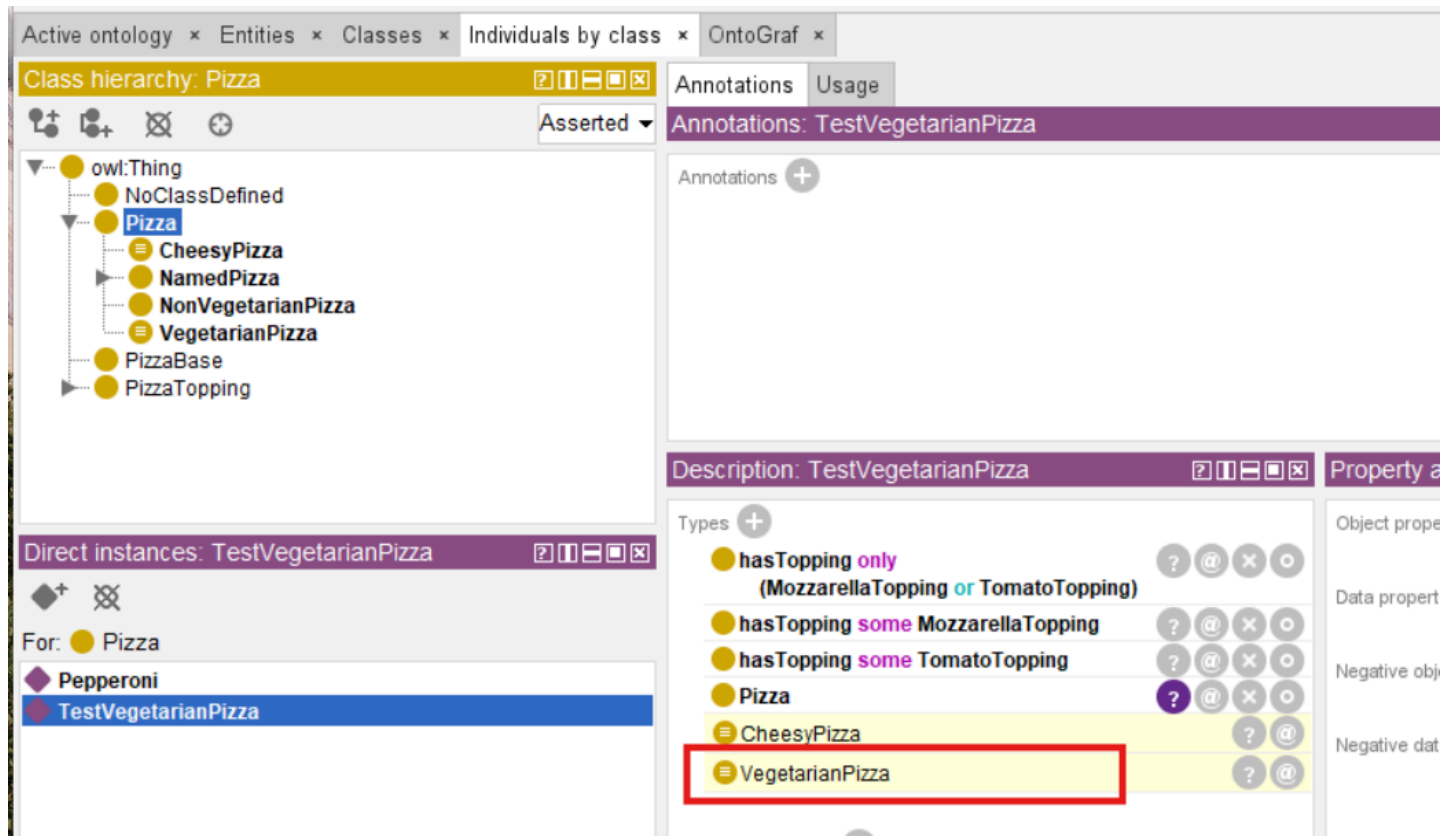
4. Click ok



Step 3: Synchronize the reasoner

Select Reasoner > Synchronize reasoner (Ctrl+R)

Step 4: Observe the result



TestVegetarianPizza is now classified as a VegetarianPizza, along with CheesyPizza. The closure axiom has resolved the same OWA problem at the individual level that it resolved at the class level.

This confirms that closure axioms are effective at both levels. In production ontologies, they are usually applied at the class level (as we did for `MargheritaPizza` and `SohoPizza`) so that every instance of the class inherits the closure. But the same principle applies to individual-level closure when a one-off case requires it.

Key Insight:

"Closure axioms are a general mechanism. They close the world for whatever they are attached to -- a class or an individual. The reasoning principle is the same."

24.6.4 From Description to Specification

Both the class-level and individual-level closure axioms have converted descriptions into specifications.

Before the closure axiom, the reasoner could say: "If you find a `MargheritaPizza` , it has mozzarella and tomato." It could not say: "If you find a pizza with mozzarella and tomato, it is a `MargheritaPizza` ."

After the closure axiom, the reasoner can say both. The ontology is now deterministic for these classes and individuals.

Key Insight:

"The closure axiom is the bridge from description to specification. Before it, the ontology describes. After it, the ontology specifies. Specification is what enables execution."

24.6.5 Why Closure Axioms Matter for Execution

Closure axioms are essential for execution because they solve the central tension between OWA and executability:

Aspect	Without Closure Axiom	With Closure Axiom
Toppings allows	Any topping not explicitly forbidden	Only the enumerated toppings
Reasoner behavior	Cannot exclude possibilities	Can exclude possibilities
Execution readiness	Not executable	Executable
Semantic strength	Description	Specification

The closure axiom is the OWL construct that most directly enables execution.

Key Insight:

"Every closed-world requirements in an execution system corresponds to a closure axiom in the ontology. Closure is the semantic foundation of executability."

24.6.6 Engineering Perspective: Exercise 23 as a Case Study

From the perspective of Protégé, **Exercise 23** involves adding a small number of axioms. It appears to be a minor modeling activity.

From the perspective of ontology engineering, however, something profound has occurred. The ontology has crossed the boundary from description to specification. It is no longer just telling us what is true -- it is telling us what is allowed.

The engineering contribution of **Exercise 23** can be summarized as follows:

Engineering Perspective	Contribution from Exercise 23
SKDL Stage	Stage 7 -- Executable Knowledge
Primary Objective	Introduce the closure axiom as the bridge from description to specification
Modeling Activity	Add closure axioms to <code>MargheritaPizza</code> , <code>SohoPizza</code> , and <code>TestVegetarianPizza</code>
Engineering Outcome	Convert descriptions into closed-world specifications; the reasoner now correctly classifies <code>MargheritaPizza</code> , <code>SohoPizza</code> , and <code>TestVegetarianPizza</code> as subclasses or instance of <code>VegetarianPizza</code>

Engineering Perspective	Contribution from Exercise 23
Preparation for Later Chapters	Establish the closure foundation for enumerated classes, <code>hasValue</code> , and cardinality restrictions

Key Insight:

"Exercise 23 is the moment the ontology becomes a specification. It is the most important step toward executability."

Last updated at 2026-09-25